Sexual reproduction in flowering plants

Flower: A Fascinating Organ of Angiosperms

1. Among the terms listed below, those that are not technically correct names for a floral whorl are
i. Androecium
ii. Carpel
iii. Corolla
iv. Sepal
1) i and iv
2) iii and iv
3) ii and iv
4) i and ii

2. In a typical complete, bisexual and hypogynous flower, the arrangement of floral whorls on the thalamus from the outermost to the innermost is
1) Calyx, corolla, androecium, and gynoecium
2) Calyx, corolla, gynoecium, and androecium
3) Gynoecium, androecium, corolla, and calyx
4) Androecium, gynoecium, corolla, and calyx

3. Flowers are objects of
(a) Aesthetic value
(b) Ornamental value
(c) Religious value
(d) Cultural value
(e) Social value
1) a and d only
2) a, b, c and e only
3) b, d and e only
4) a, b, c, d and e

4. Where does sexual reproduction take place in flowering plants?
1) Stem
2) Root
3) Flowers
4) Leaves

5. Floriculture is the study of
1) Growing and marketing flowers and foliage plants
2) Growing silkworm for production of silk
3) Growing honey bees for production of honey
4) None of the above

6. Product of sexual reproduction in angiosperms is
1) Root
2) Stem
3) Flower
4) Seed

7. The floral whorl that consists of female reproductive organs is called
1) Androecium
2) Gynoecium
3) Calyx

4) Corolla

8. (i) and (ii) constitute the accessory floral whorls whereas (iii) and (iv) constitute the reproductive whorls.

	(i)	(ii)	(iii)	(iv)
1)	Calyx	Corolla	Androecium	Gynoecium
2)	Androecium	Gynoecium	Calyx	Corolla
3)	Corolla	Androecium	Calyx	Gynoecium
4)	Gynoecium	Calyx	Corolla	Androecium

9. Flowers are absent in

(i) Monocots
(ii) Dicots
(iii) Ferns
(iv) Gymnosperms
(v) Mosses

1) (i), (iv) and (v)
2) (ii), (iii) and (iv)
3) (iii), (iv) and (v)
4) (i) and (ii) only

Pre-Fertilization: Structures and Events

10. Read the following statements and choose the correct option.

1) The number and length of stamens are variable in flowers of same species.
2) The outer three wall layers of microsporangium perform the function of protection and nourishment to the developing pollen grains.
3) Much before the actual flower is seen on a plant, the decision that the plant is going to flower has taken place.
4) Each cell of the sporogenous tissue is capable of giving rise to a macrospore tetrad.

11. A typical anther is bilobed and tetrasporangiate. It is characterized with all these features, except

1) Surrounded by four wall layers
2) Two microsporangia in each lobe
3) Sporogenous tissues at the centre of each microsporangia
4) One pollen sac in each lobe

12. Role of tapetum in pollen development is all except

1) Provides nutrition to developing pollen grains
2) Secretes sporopollenin, pollen kitt, and compatibility proteins
3) Contributes callase enzyme for separation of pollens from microspore tetrad
4) Helps in release of pollen grains

13. Select the incorrectly matched pair.

1) Endothecium -- Presence of a-cellulose fibrous bands
2) Middle layer -- Contain dense cytoplasm and more than one nucleus
3) Tapetum -- Provides nourishment to the developing microspore
4) Sporogenous -- tissue-Undergoes meiosis to form nicrospore teuad

14. Which of the following is a mismatched pair?

1) Storage of pollen grain -- In liquid N_2 at -196°C
2) Pollen allergy -- Carrot grass
3) Pollen kitt -- Insect attractant
4) Intine of pollen grain -- Sporopollenin

15. In a majority of angiosperms, the pollen grains
1) Are shed at three-celled stage
2) Carry two male gametes from beginning
3) Lack vegetative cell
4) Have one generative and one vegetative cell

16. All are features related to tapetal cells, except
1) Dense cytoplasm
2) Multinucleate
3) Polyploid
4) Have little or no DNA

17. Viability of pollen grains is maintained for months in all, except
1) Solanaceae
2) Rosaceae
3) Leguminosae
4) Poaceae

18. Select the incorrect statement with respect to Polygonumtype of embryo sac.
1) It is a 7-celled and 8-nucleated structure.
2) It shows monosporic development.
3) Arrangement of nuclei is 3 +2+3.
4) It possesses egg apparatus at the chalazal end.

19. Match the columns, and select the correct option.

	Column I		Column II
1)	Hilum	(i)	Parenchymatous tissue
2)	Micropyle	(ii)	Female gametophyte
3)	Embryo sac	(iii)	Junction between ovules and funicles
4)	Nucellus	(iv)	Small opening present at the tip of the ovule

1) a - iii; b - ii; c - iv; d- i
2) a-ii; b - iii; c - iv; d-i
3) a- iii; b - iv; c-ii; d- i
4) a- i; b - iv; c -ii; d - iii

20. Read the following statements and choose the correct option.
I. Located in the nucellus is the embryo sac or the female gametophyte.
II. Cells of the nucellus have abundant reserve food materials.
III. Integuments encircle the nucellus except at the tip, where a small opening called the hilum is organised.
IV. The number of ovules in an ovary may be one or two.
Which of the above statements are correct?
1) I and II
2) II and IV
3) I and II
4) III and IV

21. Which of the following is not a structural part of pistil?
1) Stigma
2) Ovary
3) Style
4) Connective

22. The mature female gametophyte is generally a
1) 7-celled and 7-nucleated structure

2) 8-celled and 8-nucleated structure
3) 3-celled and 3-nucleated structure
4) 7-celled and 8-nucleated structure

23. Filiform apparatus
1) Are finger-like projections
2) Direct the pollen tube growth inside the embryo sac
3) Are special cellular thickenings present at micropylar tip of synergids
4) All of the above

24. How many megaspore mother cells are required to produce 100 eggs during meiosis?
1) 100
2) 50
3) 25
4) 200

25. Egg apparatus of an embryo sac consists of
1) Egg cells only
2) One egg cell + polar nuclei
3) One egg cell + two synergids
4) One egg cell + two synergids + two polar nuclei

26. The term pollination signifies
1) Dehiscence of seeds
2) Transfer of pollen grains from anther to stigma.
3) Formation of pollinia
4) Opening of floral bud

27. Mark the odd one out with respect to the advantages of cross-pollination.
1) It increases resistance to diseases.
2) It introduces genetic variations.
3) It enhances adaptability of the offspring to change in the environment.
4) It maintains pure lines.

28. Even in the absence of absence of pollinating agents, seed setting is assured in
1) Papaya
2) Cucumber
3) Salvia
4) Commelina

29. Identify the statements that are true for anemophilous flowers.
(a) Pollen grains are light and sticky.
(b) They possess large feathery stigma.
(c) They are not very colourful and do not produce nectar.
(d) They have many ovules in each ovary.
1) (a), (b), and (c)
2) (a), (b), and (d)
3) (b) and (c)
4) (b), (c), and (d)

30. Contrivance for self-pollination/autogamy is
1) Homogamy
2) Bisexuality
3) Cleistogamy
4) All of the above

31. Go through the following points:

I. Dicliny
II. Dichogamy
III. Self-incompatibility
IV. Heterostyly

The above contrivances prevent

1) Autogamy
2) Xenogamy
3) Cross-pollination
4) Geitonogamy

32. Long, silky hair coming out of the cob of maize are

1) Meant for fruit dispersal
2) Act as insect attractant
3) Meant for protection
4) Long styles and stigma

33. Yucca is pollinated by

1) Pronuba moth
2) Colpa wasp
3) Honeybee
4) Lemur

34. Which of the following statements is correct about pollen-pistil interaction?

1) It is a static process involving pollen recognition followed by promotion or inhibition of the pollen.
2) It is a static process involving pollen recognition followed by promotion of the pollen
3) It is a dynamic process involving pollen recognition followed by inhibition of the pollen
4) It is a dynamic process involving pollen recognition followed by promotion or inhibition of the pollen

35. The part of gynoecium that determines the compatible nature of pollen is

1) Stigma
2) Style
3) Ovary
4) Secondary nucleus of embryo sac

36. Emasculation in artificial hybridization is done by removal of

1) Anthers from unisexual flower of female parent
2) Anthers from bisexual flowers of female parent
3) Stigma from unisexual flowers of male parent
4) Stigma form bisexual flower of male parent

37. Pollen-pistil interaction includes all the events from pollen deposition on stigma until the

1) Pollen tubes are formed
2) Pollen grains are formed
3) Pollen tube enters the ovule
4) Male gametes are formed

Double Fertilization

38. Match the following columns and choose the correct option.

	Column I		Column II
1)	Zygote	(i)	Involves the fusion of three haploid nuclei
2)	Primary endosperm cell	(ii)	Involves the fusion of two haploid nuclei
3)	Triple fusion	(iii)	Embryo
4)	Syngamy	(iv)	Endosperm

1) 1-c, 2-d, 3-a, 4-b
2) 1-d, 2-c, 3-a, 4-b
3) 1-c, 2-d, 3-b, 4-a
4) 1-d, 2-c, 3-b, 4-a

39. Angiosperms differ from other plants of plant kingdom in having
1) Syngamy
2) Triple fusion
3) Double fertilization
4) Both (2) and (3)

40. Read the following statements, and select the correct option.
Statement A: Angiosperms exhibit double fertilization because two fusion events occur in the same embryo sac.
Statement B: Three gametes are involved in the process of triple fusion.
1) Only (A) is correct.
2) Only (B) is correct.
3) Both (A) and (B) are correct.
4) Both (A) and (B) are incorrect.

41. In an embryo sac, the cells that degenerate after fertilisation are
1) Synergids and primary endosperm cell
2) Synergids and antipodals
3) Antipodals and primary endosperm cell
4) Egg and antipodals

42. Ploidy of endosperms will be _____, if the male and female parents are tetraploid and hexaploid, respectively.
1) 16X
2) 8X
3) 7X
4) 10X

43. Match the columns.

	Column I		Column II
1)	Primary endosperm nucleus	(i)	2n
2)	Zygote	(ii)	n
3)	Pollen grains	(iii)	3n
4)	Megaspore mother cell	(iv)	4n

1) 1-c, 2-a, 3-b, 4 – a
2) 1-c, 2-a, 3-b, 4-d
3) 1-b, 2- a, 3- c, 4-a
4) 1-b, 2-a, 3- c, 4-b

44. How many nuclei are involved in syngamy?
1) Three

2) Two
3) Five
4) Seven

45. Match the columns.

	Column I		Column II
1)	End product of sexual reproduction	a	Sporogenous tissue
2)	Occupies the centre of each microsporangium	b	Seeds and fruits
3)	Mass of cells enclosed within the integuments	c	Filiform apparatus
4)	Guides the entry of pollen tube	d	Nucellus

1) 1-b, 2-d, 3- a, 4-c
2) 1-d, 2-a, 3-b, 4-c
3) 1-b, 2 - a, 3 - d, 4-c
4) 1- b, 2 - d, 3 -c, 4-a

46. Statement I: Synergids of embryo sac help in fertilization in angiosperms.
Statement II: Filiform apparatus of synergids present at micropylar end guides the entry of pollen tube.
1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true.

47. Double fertilization was first discovered in 1898 in Lilium and Fritillaria by
1) Nawaschin
2) Strasburger
3) Amici
4) Focke

48. Identify the following statements as true (T) or false (F), and select correct option.
A. Pistil has the ability to recognize the pollen with respect to its compatibility.
B. Pollen tube discharges its male gametes on degenerating synergids.
C. Emasculation is required when female parent produces unisexual flowers.
D. Bagging prevents contamination of stigma with unwanted pollen.
1) A-T; B-F; C-F; D-T
2) A-T; B--T; C–F; D–T
3) A--F; B-F; C-T; D-F
4) A-F; B-F; C-T; D-T

Post-Fertilization: Structures and Events

49. Formation of embryo from zygote includes all except
1) Meiosis
2) Cell differentiation
3) Equational division
4) Mitosis

50. Endosperm development in angiosperms
1) Preceeds the development of embryo
2) Is mostly of free nuclear type
3) Can occur even before fertilization
4) Both (1) and (2)

51. Read the following statements and choose the incorrect option.

1) Dehydration and dormancy of mature seeds are crucial for storage of seeds.
2) Seeds can be used as food throughout the year and to raise crops in the next season.
3) The seeds remain alive only for months after they are dispersed.
4) The early stages of embryo development are similar in both monocotyledons and dicotyledons.

52. A dicot embryo consists of
1) Two cotyledons only
2) Embryonal axis and two cotyledons
3) Embryonal axis, one cotyledon and endosperm
4) Embryonal axis and scutellum

53. Which of the following is not the adaptive strategy for seed?
1) Seed cannot be stored for longer duration
2) Ensure the continuity of race from generation to generation
3) Have sufficient food reserve for young seedling
4) Being a product of sexual reproduction, they generate new genetic variations

54. Dormancy of seed is
1) State of inactivity
2) State of activity
3) Ability to germinate
4) Viability of seeds

55. Fruits of strawberry and apple resemble in
1) Parthenocarpic development
2) Contribution of thalamus in fruit formation
3) Development from multicarpellary apocarpous ovary
4) Involving no floral parts in addition to ovary

56. Tiny seeds are found in
1) Orobanche and Striga
2) Orchids
3) Ficus
4) All of the above

57. Milky water of green coconut is
1) Liquid female gametophyte
2) Liquid nucellar cells
3) Multinucleated liquid perisperm
4) Liquid endosperm

58. The oldest viable seeds are of A that germinated and flowered after an estimated record of B years
1) A-Phoenix dactylifera; B-2000
2) A-Lupinus arcticus; B–10000
3) A-Lupine; B-2000
4) A-Date palm; B-5000

59. Fill in the blanks, and choose the correct option.
(i) A is the most common type of endosperm.
(ii) The epicotyl in embryonal axis terminates into B
(iii) In grass family, the remains of second cotyledon are called C
1) A-Cellular; B-Radicle; C-Scutellum
2) A-Cellular; B-Radicle; C-Epiblast
3) A-Free nuclear; B-Plumule; C-Epiblast

4) A-Free nuclear; B-Plumule; C-Scutellum

60. The crucial feature (s) that help in storage of seeds that can be used as food throughout the year and to raise crops in next season is/are

1) Dehydration
2) Dormancy
3) Dispersal to new habitat
4) Both (1) and (2)

61. Read the following statements with respect to parthenocarpic fruits.

A. They develop without fertilization.
B. They are seedless fruits.
C. They can be produced through the application of growth hormones.

Correct statements is/are

1) Only (A)
2) Only (A)and (B)
3) Only (A) and (C)
4) All (A), (B), and (C)

62. Aleurone layer has same ploidy as that of

1) Pericarp
2) Scutellum
3) PEN
4) Coleoptile

Apomixis and Polyembryony

63. Apomixis is

1) Production of asexual seed
2) Production of sexual seed
3) Transfer of apomictic genes into hybrid varieties
4) A kind of sexual reproduction that mimics the asexual reproduction

64. What is the possible advantage if hybrids are made into apomictic?

1) Segregation of characters in the hybrid progeny
2) Absence of segregation of characters in hybrid progeny
3) Farmers can keep on using the hybrid seeds to raise new crops for many years
4) Both (2) and (3)

65. Which of the following type of embryos are apomictic?

1) Nucellar
2) Integumentary
3) Zygotic
4) Both (1) and (2)

66. The phenomenon observed in some plants wherein parts of the sexual apparatus is used for forming embryos without fertilisation is called

1) Parthenocarpy
2) Apomixis
3) Vegetative propagation
4) Sexual reproduction

67. Read the given statements.

Statement I: Hybrid seeds have to be produced every year because seeds collected from hybrid plants, if sown subsequently, do not maintain hybrid characters.

Statement II: These seeds show segregation of traits.

1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true.

68. The process of producing seed without fertilization is called

1) Parthenocarpy
2) Apomixis
3) Polyembryony
4) Parthenogenesis

69. Read the given statements.

StatementA: Apomixis is a form of asexual reproduction that mimics the sexual reproduction.

Statement B: Apomixis is common in many varieties of Citrus and Mango.

1) Only (A) is correct.
2) Only (B) is correct.
3) Both (A) and (B) are correct.
4) Both (A) and (B) are incorrect.

70. Match the columns I and II and choose the correct combination from the options given.

	Column I		Column II
a	Apomixis	1	Occurrence of more than one embryo in seed
b	Polyembryony	2	Seedless fruit
c	Parthenocarpy	3	Seed without fertilisation

	A	B	C
1)	3	1	2
2)	2	3	1
3)	1	2	3
4)	3	2	1

71. Polyembryony is the

1) Occurrence of more than one seed in an embryo
2) Occurrence of more than one embryo in a seed
3) Occurrence of more than one seed in fruit
4) Occurrence of more than one embryo in a fruit

72. Cultivation of hybrids has tremendously increased productivity. The disadvantage of hybrid seed is

1) Hybrid seeds have to be produced every year
2) Production of hybrid seeds is costly
3) If the seeds collected from hybrids are sown, the plants in the progeny will segregate and do not maintain hybrid characters
4) All of the above

73. Read the following statements and choose the incorrect option.

1) Apomixis is a form of asexual reproduction that mimics sexual reproduction.
2) Seeds are the products of asexual reproduction.
3) Apomicts have several advantages in horticulture and agriculture.
4) All the above

Answer Key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	1	4	3	1	4	2	1	3	3	4	4	2	4	4
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	4	4	3	3	4	4	4	1	3	2	4	4	3	4
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
1	4	1	4	1	2	3	1	4	1	2	2	1	2	3
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
1	1	2	1	4	3	2	1	1	2	4	4	2	3	4
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**	**71**	**72**	**73**		
4	3	1	4	4	2	1	2	3	1	2	4	2		